

Radioactive Duck

Student's name	: Morard Ambre, Tornare Lucas, Contat Sonny
Module	: 64-31 Web Development
Submitted on 25.06.2026	

Table of contents

1	Indications of the integrated competencies and their location in the code	2
2	Links	5

1 Indications of the integrated competencies and their location in the code

Skills	Mandatory project implementation	Indicate file and line + screenshot
Forms HTML elements	x (form with HTML input and validation see optional JS)	<i>Index.html (l. 42-43)</i> <i>Js/setup.js (l. 14-22)</i>
	<pre> 39 <!-- 1) FORMULAIRE : pseudo + validation --> 40 <div class="setup-section"> 41 <h3>1. Your nickname</h3> 42 <input type="text" id="pseudo" maxlength="15" 43 placeholder="2 to 15 characters (letters, numbers, _ or -)" autocomplete="off"> 44 <div id="pseudo-error"></div> 45 </div> </pre> <pre> 13 // Check the nickname: it must be 2 to 15 characters long 14 function validatePseudo() { 15 const value = pseudoInput.value.trim(); 16 const ok = value.length >= 2 && value.length <= 15; 17 pseudoError.textContent = (value.length > 0 && !ok) 18 ? 'Nickname must be 2 to 15 characters long.' 19 : ''; 20 return ok; 21 } 22 pseudoInput.addEventListener('input', validatePseudo); </pre>	
Medias & API	x (canvas AND audio or video e.g.: integrates a tutorial video of the game or application)	<i>Game.html (l. 26)</i> <i>js/audio.js (l.1-10)</i>
	<pre> <canvas id="gameCanvas" width="800" height="600"></canvas> </pre> <pre> 1 const birdDeath = new Audio('ressources/audio/birdDeath.mp3'); 2 const gunFire = new Audio('ressources/audio/gunFire.mp3'); 3 gunFire.volume = 0.2; // lower volume of gun fire sound 4 const ratDeath = new Audio('ressources/audio/ratDeath.mp3'); 5 const foxDeath = new Audio('ressources/audio/foxDeath.mp3'); 6 const playerDeath = new Audio('ressources/audio/playerDeath.mp3'); 7 const musicGameMenu = new Audio('ressources/audio/musicGameMenu.mp3'); 8 const musicGameplay = new Audio('ressources/audio/musicGameplay.mp3'); 9 musicGameMenu.loop = true; 10 musicGameplay.loop = true; </pre>	

Skills	Mandatory project implementation	Indicate file and line + screenshot
DOM handling functions	x (min. getElementById() or querySelector())	js/setup.js (l.4-8, 31)
	<pre> 3 // DOM references 4 const pseudoInput = document.getElementById('pseudo'); 5 const pseudoError = document.getElementById('pseudo-error'); 6 const duckError = document.getElementById('duck-error'); 7 const stage = document.getElementById('avatar-stage'); 8 const playBtn = document.getElementById('play-btn'); 30 // Each duck can be dragged into the drop zone, or simply clicked 31 document.querySelectorAll('#duck-choices .duck-option').forEach((opt) => { 32 opt.addEventListener('dragstart', (e) => e.dataTransfer.setData('text/plain', opt.dataset.value)); 33 opt.addEventListener('click', () => chooseDuck(opt.dataset.value)); 34 }); </pre>	
Several variable scopes in JS	x (implicit)	js/game.js (l. 2-4, 16-17)
	global <pre> 2 let gameStarted = false; 3 let gamePaused = false; 4 let gameOver = false; 5 let showHallofFame = false; </pre> VS local <pre> 17 const frameW = 138; 18 const frameH = 138; </pre>	
Natives Libraries	x (min. Math)	js/projectile.js (l.37, 48) js/game.js (l.21)
	<pre> 37 const length = Math.sqrt(dx * dx + dy * dy); 48 angle: Math.atan2(dy, dx), 21 const row = Math.floor(player.frameX / 2); </pre>	

Skills	Mandatory project implementation	Indicate file and line + screenshot
Variable comparison operators	x (implicit)	js/player.js (l. 5-14)
	<pre> 4 // Redirect to the customization page if the player is not configured correctly 5 if (6 !savedPlayer 7 !savedPlayer.pseudo 8 savedPlayer.pseudo.trim().length < 2 9 savedPlayer.pseudo.trim().length > 15 10 !['yellow', 'pink', 'green'].includes(savedPlayer.duck) 11) { 12 window.location.href = 'index.html'; 13 throw new Error('Player not configured'); 14 }</pre>	
Loop & iterators	x (loop on object)	js/projectile.js (l.72-82)
	<pre> 72 function drawProjectiles() { 73 for (const projectile of projectiles) { 74 if (projectile.owner === "player") { 75 drawFireball(playerFireballFrames, projectile); 76 } else if (projectile.owner === "crow") { 77 drawFireball(crowFireballFrames, projectile); 78 } else if (projectile.owner === "fox") { 79 drawFireball(foxFireballFrames, projectile); 80 } 81 } 82 };</pre>	
Types of functions: - standard, - anonymous	x (standard function with parameters and method)	js/enemy.js (l. 65) js/input-handler.js (l.8)
	<pre> 65 function spawnEnemies(waveNumber) { 8 canvas.addEventListener('mousemove', (event) => {</pre>	
Types de stockages (bonus)	x (local storage)	js/setup.js (l. 63) js/player.js (l.64-74)
	<pre> 63 localStorage.setItem('duckPlayer', JSON.stringify(player));</pre>	

Skills	Mandatory project implementation	Indicate file and line + screenshot
	<pre> 61 // Persists the run's score to localStorage so it survives page reloads/sessions. 62 // Called from restartPlayer(), right before player.score is reset to 0. 63 function scoreSaver() { 64 let bestScore = localStorage.getItem("bestScore"); 65 if (bestScore === null player.score > parseInt(bestScore)) { 66 localStorage.setItem("bestScore", player.score); 67 } 68 localStorage.setItem("lastScore", player.score); 69 }</pre>	
Callback	x (either by use of API or own construction)	js/game.js (l. 330)
	<pre> 339 requestAnimationFrame(gameLoop);</pre>	
Drag'n Drop	x (Building an avatar)	index.html (l.52-67) js/setup.js (l.31-34)
	<pre> <div class="duck-option" draggable="true" data-value="yellow"></pre> <pre> 30 // Each duck can be dragged into the drop zone, or simply clicked 31 document.querySelectorAll('#duck-choices .duck-option').forEach((opt) => { 32 opt.addEventListener('dragstart', (e) => e.dataTransfer.setData('text/plain', opt.dataset.value)); 33 opt.addEventListener('click', () => chooseDuck(opt.dataset.value)); 34 });</pre>	
JSON	OPTION (Game setup et React JS website)	js/setup.js (l. 63) js/player.js (l. 2)
	<pre> 63 localStorage.setItem('duckPlayer', JSON.stringify(player));</pre> <pre> 1 // Get the player configuration saved from the customization page 2 const savedPlayer = JSON.parse(localStorage.getItem('duckPlayer') 'null');</pre>	

2 Links

- Link to source code (game):
https://github.com/LucasTornare/64-31_RadioactiveDuckGame
- Link to source code (react):
https://github.com/LucasTornare/64-31_webdevelopment_react_group_b

- Link to demo: https://lucastornare.github.io/64-31_RadioactiveDuckGame/
- Link to website:
<https://radioactive-duck-fdacdkdmbfsfndged.switzerlandnorth-01.azurewebsites.net/description>